

# Sprint 4 Report

Hyper-Localized Multilingual Voice Assistant

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**Sprint:** 4 of 6    **Duration:** Week 7 – 8    **Team:** Dev (Solo)    **Velocity:** 18 / 18 pts

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## 1. Sprint Goal

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Agentic capabilities, monitoring dashboard, database, and testing infrastructure. Build LangGraph-based agent orchestration with intent routing, integrate ClawGo/OpenClaw for local inference, add SQLite conversation logging, create a real-time monitoring dashboard, and establish a comprehensive pytest test suite.

## 2. Sprint Outcome

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All 18 story points delivered. The assistant now has intelligent agent routing, persistent storage, a live dashboard, and proper test coverage.

### Key Deliverables

- LangGraph stateful workflow — intent-based routing to Info, Task, Chat, and Smart Home agents
- ClawGo/OpenClaw Pi agent — with Ollama for local Qwen3 inference (BYOM)
- AgentFactory — runtime switching between LangGraph and ClawGo backends
- SQLite database — conversation/message/metric models for persistent logging
- Real-time dashboard — stats, latency breakdown, conversation logs, language distribution, live testing
- Dashboard auto-served at root (/) when API starts
- Comprehensive test suite — conftest.py fixtures, core tests, DB CRUD tests, API endpoint tests
- Repository cleanup — removed 13 unnecessary files

### 3. Scrum Status Report

| Task Description                                   | Date | Completion Status | Comments                                       |
|----------------------------------------------------|------|-------------------|------------------------------------------------|
| T4.1 — LangGraph agent orchestrator (3 pts)        |      | Completed         | Stateful workflow with intent-based routing    |
| T4.2 — ClawGo/OpenClaw agent + Ollama BYOM (3 pts) |      | Completed         | Local Qwen3 inference via Ollama               |
| T4.3 — Agent factory routing (2 pts)               |      | Completed         | Runtime switching: LangGraph / ClawGo          |
| T4.4 — SQLite database for logging (2 pts)         |      | Completed         | Conversation/message/metric models             |
| T4.5 — Frontend dashboard (3 pts)                  |      | Completed         | Stats, latency, conversations, live test panel |
| T4.6 — Dashboard API endpoints (2 pts)             |      | Completed         | Stats + conversations endpoints; served at /   |
| T4.7 — Comprehensive test suite (3 pts)            |      | Completed         | conftest, test_basic, test_database, test_api  |

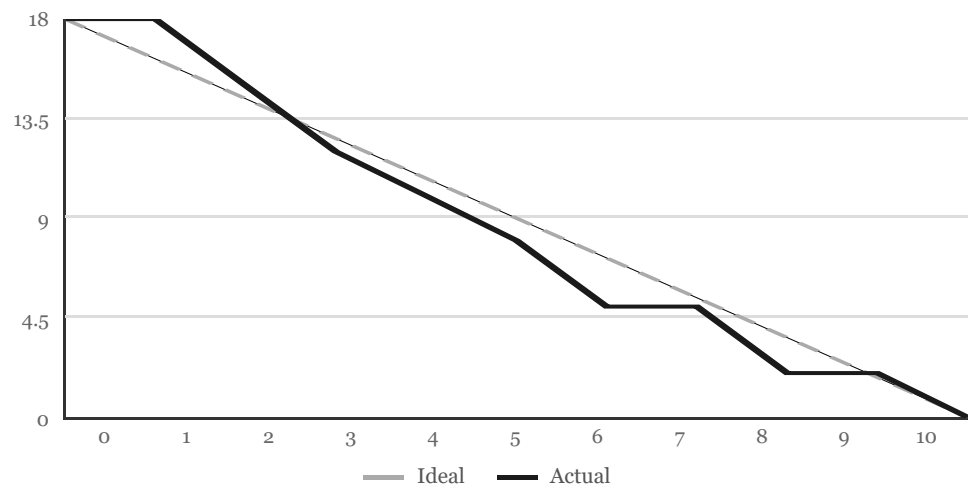
Total: 7 tasks, 18 story points, 100% complete.

## 4. Daily Scrum Meeting Log

| Date | Work Done                                                                             | Work Planned                                                                        | Issues                                                     | Sign |
|------|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------|------|
|      | Sprint 3 retrospective; reviewed pipeline, API, model upgrade deliverables            | Design LangGraph agent orchestrator architecture (T4.1); define state schema        | None                                                       |      |
|      | Designed agent graph: classify_intent → route → [info task chat smart_home] → respond | Implement LangGraph orchestrator with stateful graph (T4.1); test intent routing    | None                                                       |      |
|      | LangGraph orchestrator working with 4 agent types (T4.1 done)                         | Research ClawGo/OpenClaw framework (T4.2); set up Ollama for local Qwen3            | None                                                       |      |
|      | ClawGo agent integrated with Ollama BYOM; Qwen3 running locally                       | Complete ClawGo testing (T4.2); build AgentFactory for backend switching (T4.3)     | Ollama model download slow — resolved with pre-pull script |      |
|      | ClawGo complete (T4.2 done); AgentFactory config-based routing (T4.3 done)            | Design SQLite schema for conversations/messages/metrics (T4.4); implement CRUD      | None                                                       |      |
|      | SQLite database with conversation logging complete (T4.4 done)                        | Begin frontend dashboard — layout, stats cards, latency chart (T4.5)                | None                                                       |      |
|      | Dashboard layout with stats row, latency bars, conversation list done                 | Add language distribution + Try It panel (T4.5); dashboard API endpoints (T4.6)     | None                                                       |      |
|      | Dashboard feature-complete (T4.5 done); API endpoints serving data (T4.6 done)        | Set up pytest infrastructure — conftest.py, test structure (T4.7); write core tests | None                                                       |      |
|      | Test fixtures + core tests written; database CRUD tests added                         | Write API endpoint tests (T4.7); clean up repository — remove 13 files              | None                                                       |      |
|      | API tests complete; all tests passing; repo cleanup done                              | Full test suite verification (T4.7 done); sprint wrap-up; update SCRUM_BOOK.md      | None                                                       |      |

## 5. Burndown Chart

18 story points over 10 working days



### Burndown Data

| Day | Ideal | Actual | Done | Notes                                |
|-----|-------|--------|------|--------------------------------------|
| 0   | 18.0  | 18     | 0    | Sprint start                         |
| 1   | 16.2  | 18     | 0    | T4.1 architecture design             |
| 2   | 14.4  | 15     | 3    | T4.1 LangGraph orchestrator complete |
| 3   | 12.6  | 12     | 6    | T4.2 ClawGo agent in progress        |
| 4   | 10.8  | 10     | 8    | T4.2 + T4.3 complete                 |
| 5   | 9.0   | 8      | 10   | T4.4 SQLite database complete        |
| 6   | 7.2   | 5      | 13   | T4.5 dashboard in progress (ahead)   |
| 7   | 5.4   | 5      | 13   | T4.5 + T4.6 completion               |
| 8   | 3.6   | 2      | 16   | T4.5 + T4.6 done; tests started      |
| 9   | 1.8   | 2      | 16   | T4.7 API tests in progress           |
| 10  | 0.0   | 0      | 18   | T4.7 complete; sprint done           |

## 6. Next Sprint Backlog — Sprint 5

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**Goal:** Running on Raspberry Pi 5 with optimized performance.

| ID   | Task                   | Assignee | Points | Priority |
|------|------------------------|----------|--------|----------|
| T5.1 | Pi 5 environment setup | Dev      | 2      | High     |
| T5.2 | Model quantization     | Dev      | 3      | High     |
| T5.3 | ONNX conversion        | Dev      | 3      | High     |
| T5.4 | Audio I/O on Pi        | Dev      | 3      | High     |
| T5.5 | Response caching       | Dev      | 2      | Medium   |
| T5.6 | Performance profiling  | Dev      | 2      | Medium   |
| T5.7 | Latency optimization   | Dev      | 3      | High     |

Total: 7 tasks, 18 story points.